

① Queue Data Structure :-

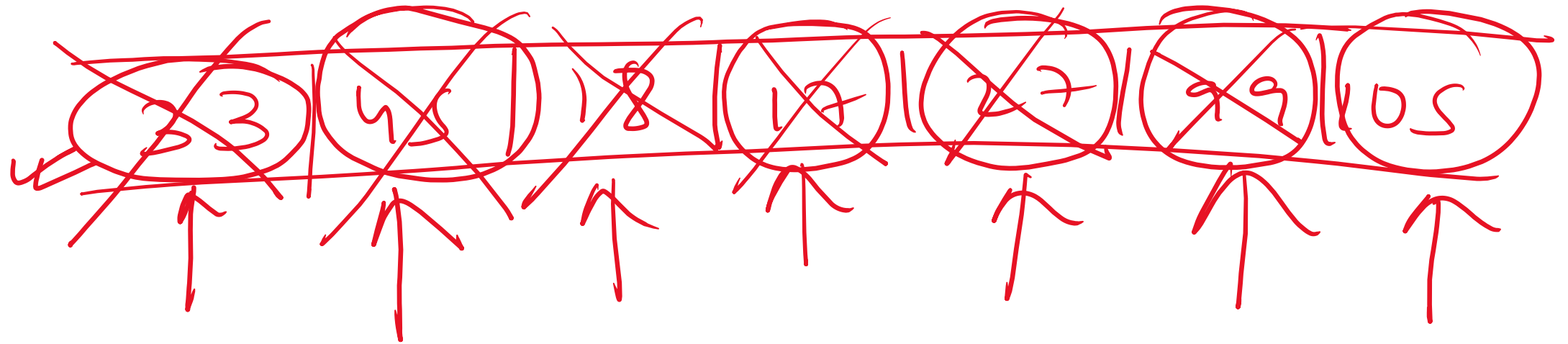
1. It follows the logic of FIFO
(First In First Out).

that implies the element which has been inserted first into queue will be the first element to get deleted from queue.

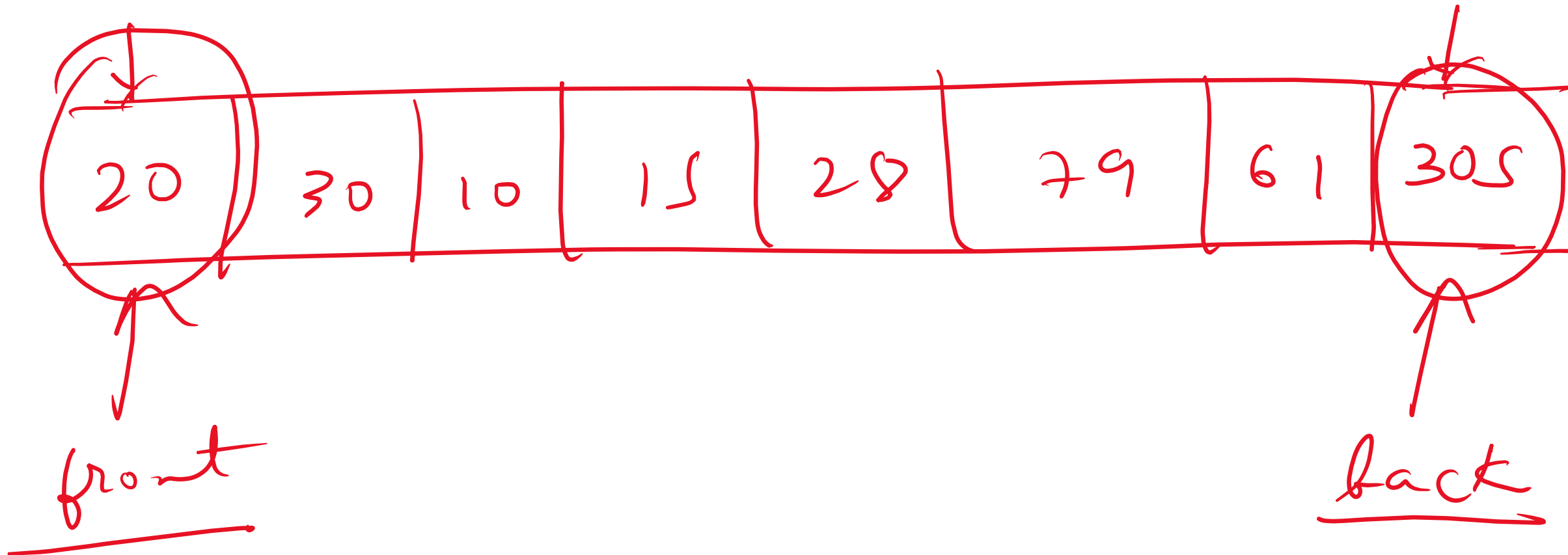
o Array Implementation

o Linked List ' ' .

eg:- 33, 45, 18, 17, 27, 99, 105



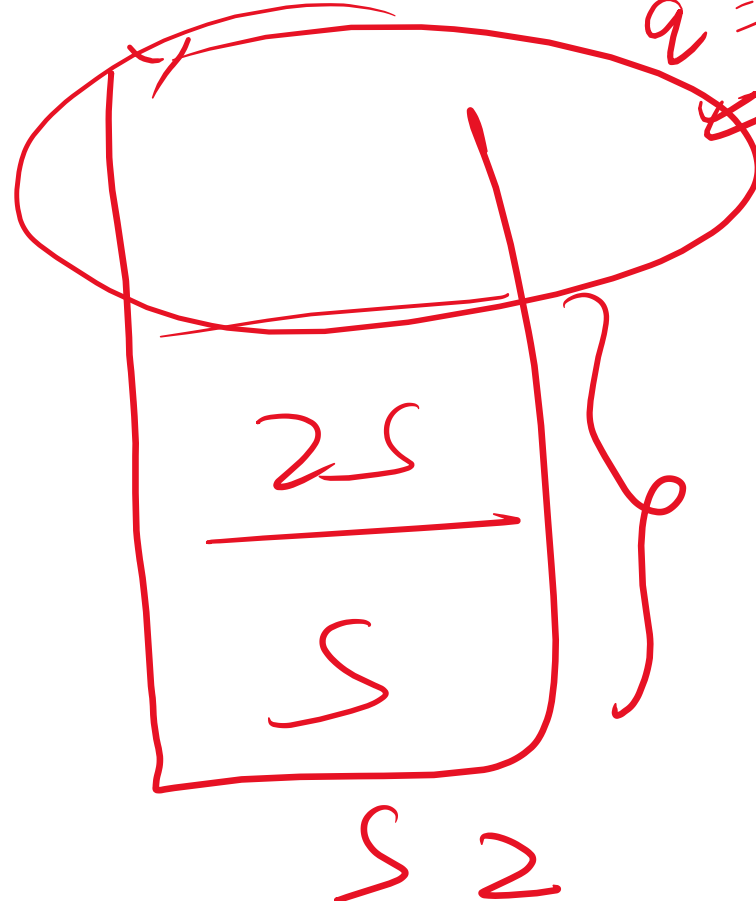
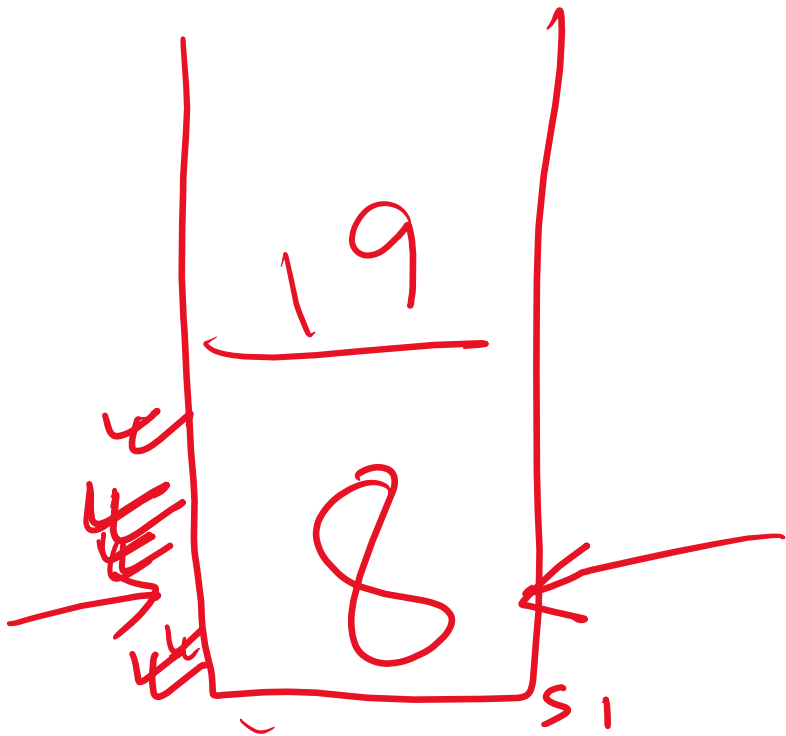
① Deque (Doubly Ended Queue) :-



eg:- 15, 25, 5, 8, 19, 27, 92

q = { 15, 25, 5, 8, 19, 27, 92 }

q = 25, 5, 8, 19



FIFO

LIFO

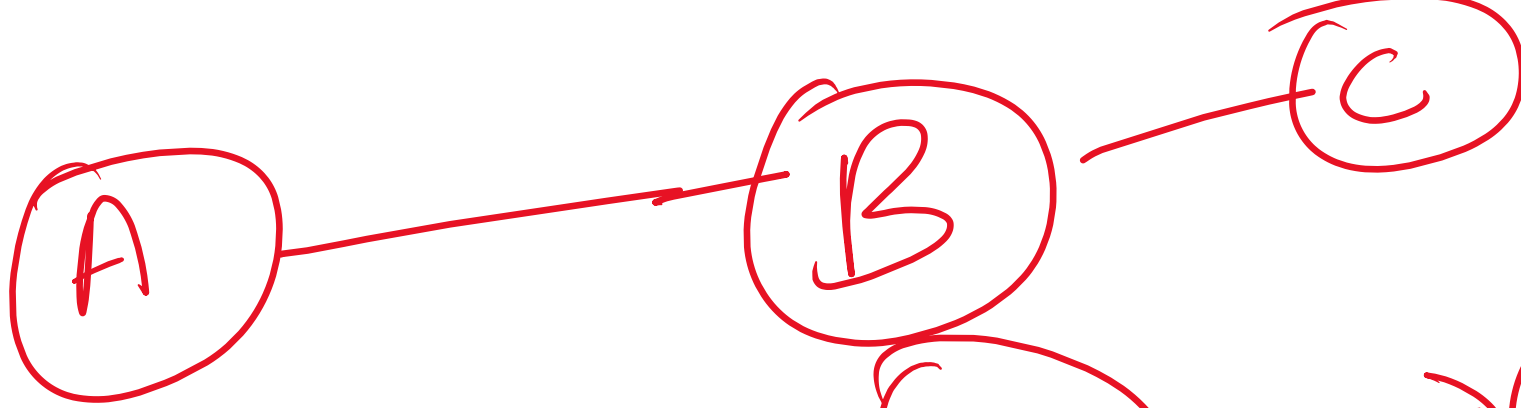
① Logic :-

1. Start from the head Node and iterate through each Node and if any Node has child pointer, push that child pointer Node into the queue.

2. Once you reach the Tail Node
or a node with null next pointer then
of any level_n extract the front
element from the queue bcoz
that is the door to move to the
Next level and attach that
front element of the queue with
the tail Node of previous level.

3. Tail element is that element which has neither next pointer nor child pointer.

UG:-



DG:-

